

PRODUCT VISION

Enterprise Data Lake Platform

DRAFT V1

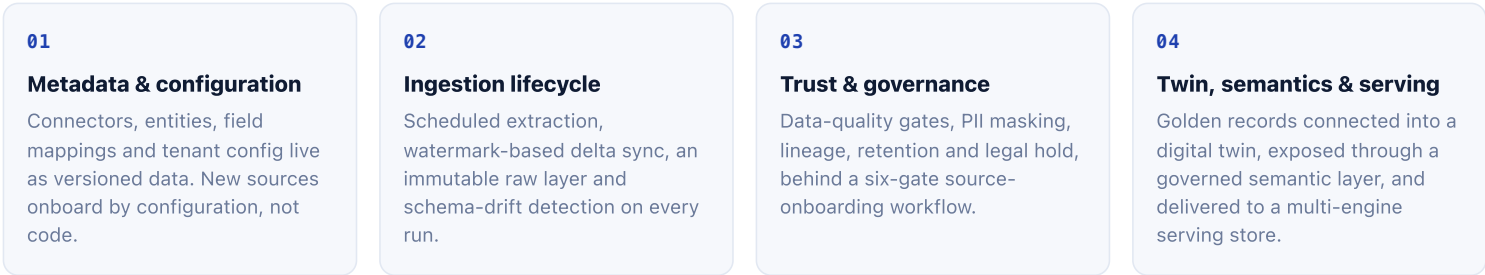
A governed, metadata-driven pipeline that turns siloed source systems into one trusted, entity-resolved analytics foundation — connected into a **digital twin**, exposed through a governed **semantic layer**, and paired with a self-service control plane that lets each tenant configure it without engineering.

THE BACKBONE

DL

The Governed Pipeline

One metadata-driven backbone carries every record from source system to trusted analytics — extraction, quality, resolution, the digital twin, a semantic layer and serving — tenant-scoped and fully audited from end to end.



SS

Self-Service Configuration
Tenant users configure connectors, mappings, resolution rules, schedules and serving stores through a permission-gated web console — with draft/publish, dry-run validation and maker-checker approval. Engineering is no longer in the loop for onboarding.

FROM SOURCE TO INSIGHT — THE RECORD LIFECYCLE



MODULAR PRODUCT MODEL

Composable modules on one pipeline

Each module is an independently deployable part of the same metadata-driven platform — sharing one tenant model, one versioned configuration system and one audit trail.

CR

Connector Runtime & Orchestration Backbone

Adapter-and-registry source connectors (Salesforce, MySQL RDS, NetSuite, Sage Intacct, Sage X3) driven by EventBridge → SQS FIFO → Step Functions → Lambda, with watermark delta sync and an immutable raw layer.

T

Transformation & Data Quality

Versioned field mapping, type-casting, PII masking, quality gates (null/range/pattern/enum) and SCD Type-1 merge.

R

Entity Resolution & Golden Records

Deterministic and probabilistic matching with config-driven survivorship, field provenance and stable golden IDs.

S

Serving Store

Five engines — MySQL, PostgreSQL, SQL Server, Azure SQL, Redshift — with idempotent upsert and per-tenant isolation.

G

Governance & Observability

Six-gate onboarding, classification, lineage, retention and legal hold; structured logs, metrics, tracing and a DLQ with replay.

C

Control Plane & Console

Cognito/JWT API plus a self-service React console — draft/publish, dry-run validation, maker-checker approval and monitoring.

TW

Digital Twin

Connects golden records into one entity with its relationships, lifecycle and rollups — a 360° view per company, contract and supplier.

SM

Semantic Layer

Governed metrics, dimensions and joins over the golden layer, so every tool and query computes the same numbers the same way.

AI

Conversational Agent

Natural-language questions become governed answers through a verification loop, with saved, re-runnable named queries.

DB

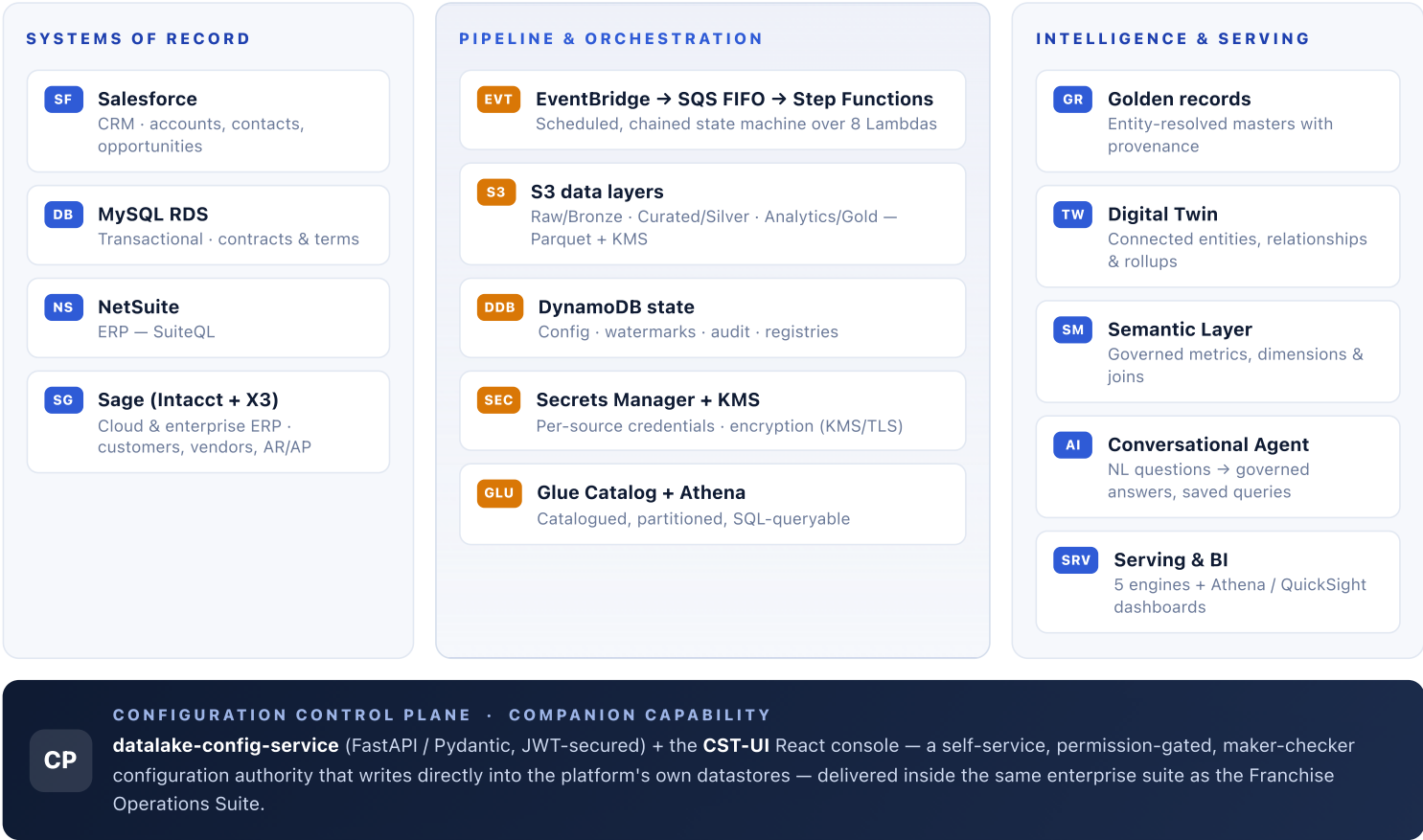
Dashboards & Reporting

Tenant-scoped, permission-aware dashboards built on saved semantic queries — one consistent set of governed definitions.

ENTERPRISE ARCHITECTURE

One pipeline. One control plane. One intelligence layer.

Serverless AWS services carry data from systems of record through governed layers to analytics and serving, while a self-service control plane configures every tenant.



PLATFORM STATUS & ROADMAP

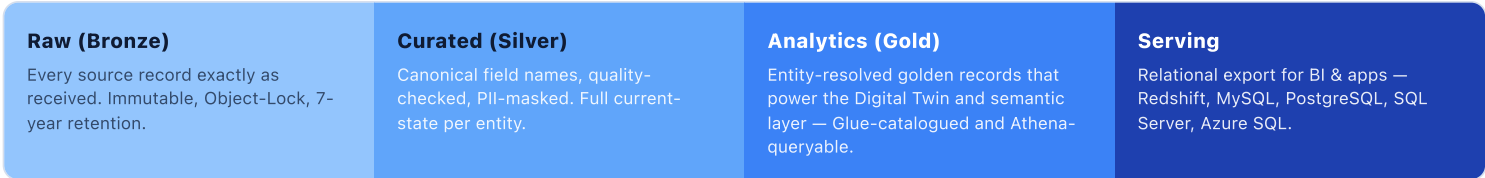
Where the platform stands — and where it's going

Dev is deployed and the pipeline runs end-to-end for its two connected sources. Staging and production are not yet provisioned. Figures below reflect the Dev environment — not a production benchmark.



SOURCE	TYPE	STATUS	ENTITIES	EXTRACTION
Salesforce	CRM	Live	Account (34), Contact, Opportunity, Contract	Bulk API 2.0 · incremental
MySQL RDS	Transactional	Live	Contracts (36,023), Contract Terms	JDBC · incremental
Sage Intacct	Cloud ERP	Ready	Customer, Vendor, AR Invoice, AP Bill	REST / OAuth 2.0
Sage X3	Enterprise ERP	Ready	Customer, Supplier	OData v4 / OAuth 2.0
NetSuite	ERP	Ready	Customer	SuiteQL REST

THE FOUR DATA LAYERS



THE PLAN AHEAD

